

Corollary 31. *Fix $m, n \in \mathbb{N}$. Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^+$ be an even C^1 function with $\psi, |\nabla\psi|^2 \in L^1(\mathbb{R}^n, \gamma_n^{\frac{m}{m+1}})$ and $D^m\psi, |\nabla D^m\psi|^2 \in L^1(\mathbb{R}^{nm}, \gamma_{n,m})$. Then,*

$$\begin{aligned} & \left(\frac{m+1}{m} \right) \text{Var}_{\gamma_n^{\frac{m}{m+1}}} \psi + \frac{1}{m+1} \text{Var}_{\gamma_{n,m}} D^m\psi \\ & \leq \frac{1}{2} \left(\frac{1}{m+1} \int_{\mathbb{R}^{nm}} |\nabla D^m\psi(x)|^2 d\gamma_{n,m}(x) + \int_{\mathbb{R}^n} |\nabla\psi(x)|^2 d\gamma_n^{\frac{m}{m+1}}(x) \right). \end{aligned}$$

Observe that $\nabla\psi$ refers to taking the gradient of ψ on $\mathbb{R}^n$ while $\nabla D^m\psi$ contains the gradient on $\mathbb{R}^{nm}$. The quantity $|\nabla D^m\psi|^2$ seems a bit mysterious, but in fact it is not so complicated. Since ∇ respects the product structure of $\mathbb{R}^{nm}$, one has that

$$\nabla D^m\psi(x_1, \dots, x_m) = \left\{ \nabla\psi(x_i) - \nabla\psi \left(-\sum_{i=1}^m x_i \right) \right\}_{i=1}^m.$$

Thus,

$$|\nabla D^m\psi(x_1, \dots, x_m)|^2 = \sum_{i=1}^m \left| \nabla\psi(x_i) - \nabla\psi \left(-\sum_{i=1}^m x_i \right) \right|^2.$$

Proof. Using standard approximation techniques, and the fast decay of the densities of the measures $\gamma_n^{\frac{m}{m+1}}$ and $\gamma_{n,m}$, we may assume that ψ is compactly supported. In Theorem 4, write $f = e^{-V}$ and switch the roles of V and V^* . Then, one can take logarithm to obtain

$$\begin{aligned} (45) \quad & m \log \left(\int_{\mathbb{R}^n} e^{-\frac{m+1}{m} V^*(x)} dx \right) + \log \left(\int_{\mathbb{R}^{nm}} e^{-D^m(V(x))} dx \right) \\ & \leq m \log \left(\int_{\mathbb{R}^n} e^{-\frac{m+1}{m} \frac{|x|^2}{2}} dx \right) + \log \left(\int_{\mathbb{R}^{nm}} e^{-D^m \frac{|x|^2}{2}(x)} dx \right). \end{aligned}$$

We consider the case when $V(x) = V_\varepsilon(x) = \frac{|x|^2}{2} + \varepsilon\psi(x)$. Focusing on the second integral in (45), we have

$$\begin{aligned} & \log \left(\int_{\mathbb{R}^{nm}} e^{-D^m(V_\varepsilon(x))} dx \right) - \log \left(\int_{\mathbb{R}^{nm}} e^{-D^m \frac{|x|^2}{2}(x)} dx \right) \\ & = \log \left(\int_{\mathbb{R}^{nm}} \exp(-\varepsilon D^m(\psi(x))) d\gamma_{n,m}(x) \right) \\ & = \log \left(1 - \varepsilon \int_{\mathbb{R}^{nm}} \left(D^m\psi(x) + \frac{\varepsilon^2}{2} (D^m\psi(x))^2 \right) d\gamma_{n,m}(x) + o(\varepsilon^3) \right) \\ & = -\varepsilon \int_{\mathbb{R}^{nm}} D^m\psi(x) d\gamma_{n,m}(x) + \frac{\varepsilon^2}{2} (\text{Var}_{\gamma_{n,m}} D^m\psi) + o(\varepsilon^3). \end{aligned}$$

We next need the well-known fact that if $\psi \in C^1(\mathbb{R}^n)$ is compactly supported, then

$$V_\varepsilon^* = \frac{|x|^2}{2} - \varepsilon\psi(x) + \frac{\varepsilon^2}{2} |\nabla\psi|^2 + o(\varepsilon^3),$$